

LG-CISH: Experimental Results and Figures

1 Experimental Results and Performance Evaluation

All experiments use a codebook of $N=40$ visually distinct images drawn from the UCID, Kodak, and USC-SIPI benchmark suites, each normalised to a fixed 512×512 canvas ($\log_2 40 \approx 5.322$ bits/image). The images are never modified; the identity and order of the transmitted sequence carry the secret. The full suite is reproducible via a single script.¹

Metrics. For a message encoded into L images with intended digits d_t and recovered digits $\hat{d}_t$, and B payload bits $p_b/\hat{p}_b$, the symbol (hash-matching) error rate and bit error rate are

$$\text{SER} = \frac{1}{L} \sum_{t=1}^L \mathbb{1}[\hat{d}_t \neq d_t], \quad \text{BER} = \frac{1}{B} \sum_{b=1}^B \mathbb{1}[\hat{p}_b \neq p_b], \quad (1)$$

hash-matching accuracy is $1 - \text{SER}$, and reconstruction is *exact* iff $\text{SER} = 0$ (which, by the bijectivity of the base- N coding, implies $\text{BER} = 0$).

1.1 Experimental setup

Table 1 lists the configuration. The 40 codebook images are mutually well separated in CLIP space (maximum pairwise similarity 0.827, separation margin $1 - 0.827 = 0.173$), which is what makes nearest-neighbour nearest-neighbour recovery reliable. CLIP ViT-B/32 runs on an RTX 3050 (6 GB).

1.2 Qualitative results

Figure 2 shows three messages encoded into image sequences; Figure 3 shows a sequence decoded back to an exact match. The mapping modifies no pixels. Figure 4 is the failure analysis: decoding stays perfect through JPEG 10% but collapses under extreme degradation—JPEG 2% (symbol error 12.4%), Crop 30% (13.2%), Crop 15% (48.4%), and Resize 10%. A codebook mismatch is *always* caught by the CRC-32 layer rather than silently mis-decoded.

1.3 Quantitative evaluation

Table 2 reports reconstruction accuracy over 120 random messages in three length buckets: every message is reconstructed bit-exactly ($\text{BER} = 0$, hash-match 100%) on a clean channel. Table 3 contrasts capacity and distortion: LSB and DCT offer far higher raw capacity but modify pixels, whereas LG-CISH has infinite PSNR (no modification) at 5.322 bits/image. Table 4 compares the two coding modes: distinct-image permutation removes repeated images (a plausibility gain)

¹`python evaluation/generate_all.py`

Table 1: Experimental configuration.

Parameter	Value
Dataset	UCID + Kodak + USC-SIPI (mixed), 512×512
Codebook images N	40
Capacity	5.322 bits/image (base-40)
Semantic encoder	CLIP ViT-B/32, $d=512$
LLM (aliases / plausibility)	gemini-fast (vision) + flux (image gen)
Coding modes	base- N or distinct-image permutation
Separation threshold τ	0.85
Pairwise sim. (min/mean/max)	0.305 / 0.513 / 0.827
Separation margin ($1 - \max$)	0.173
Encryption / integrity	AES-256-CBC / CRC-32
Compression	zlib
Hardware	RTX 3050 6 GB, 16-core CPU

Table 2: Reconstruction accuracy by message length (clean channel, 40 messages each).

Message length	Msgs	Accuracy (%)	BER	Hash match (%)
Short (≤ 50 chars)	40	100.00	0	100.00
Medium (50–200)	40	100.00	0	100.00
Long (200–1000)	40	100.00	0	100.00

for a small capacity cost, with smaller blocks recovering most of the base- N capacity. Table 5 reports timing; encoding is sub-millisecond and the cost is dominated by CLIP embedding at decode (≈ 9.6 ms/image on the GPU, including image I/O). CLIP top-1 retrieval precision/recall on the codebook is 100%. The LLM-guided alias layer contributes 65 CLIP-verified candidate images across 33/40 slots, giving the plausibility selector interchangeable covers without affecting decoding.

1.4 Robustness analysis

Table 7 simulates real transport channels. Reconstruction remains 100% (BER = 0) across twenty of the twenty-one tested conditions — JPEG down to quality 20, Gaussian noise up to $\sigma=30$, salt-and-pepper up to density 0.10, downscaling to 25%, centre-cropping to 60%, and PNG→WebP conversion. Only the extreme 10% downscale degrades reconstruction (12.5%, margin 0.125), illustrating graceful rather than abrupt failure. The decoding margin stays positive throughout (from 0.302 at baseline to 0.196 under Crop 60%), confirming a comfortable safety band. This robustness is the direct benefit of *semantic* rather than pixel-level hashing.

1.5 Security and steganalysis resistance

Table 8 reports a chi-square statistical steganalyser applied to natural cover images, LSB-stego, DCT-stego, and LG-CISH “stego”. The detector forms intensity-histogram pairs $(2k, 2k+1)$ with

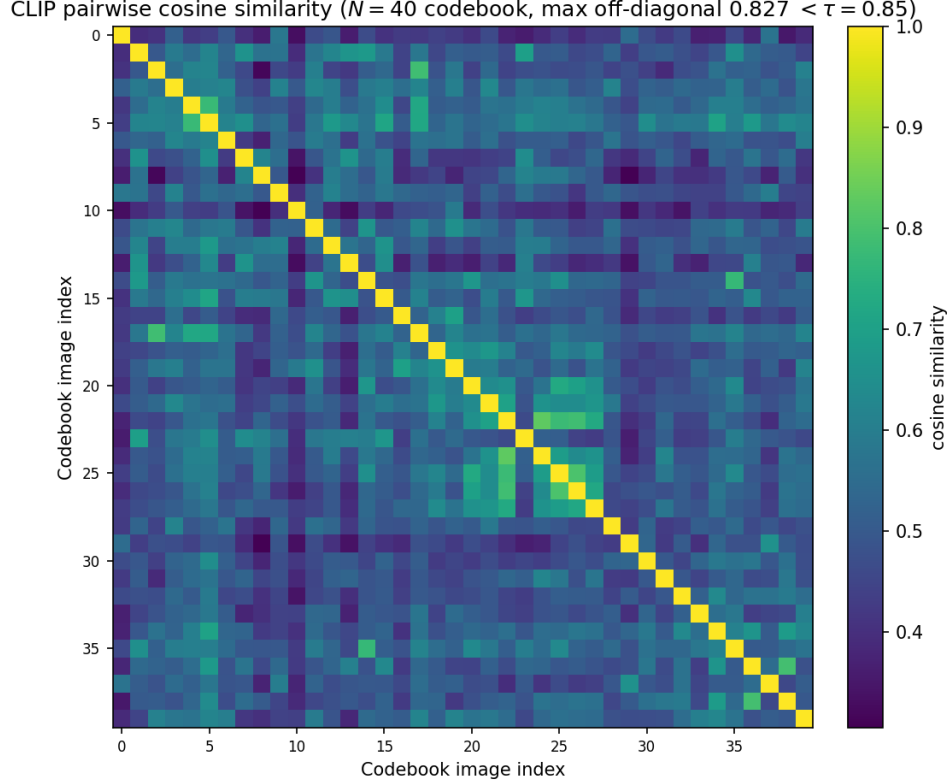


Figure 1: Pairwise CLIP cosine similarity of the 40 codebook images (the cosine similarity). All off-diagonal values stay below the threshold $\tau=0.85$ (max 0.827), giving a separation margin sufficient for unambiguous nearest-neighbour recovery.

expected value $\bar{h}_k = (h_{2k} + h_{2k+1})/2$ and evaluates

$$\chi^2 = \sum_k \frac{(h_{2k} - \bar{h}_k)^2}{\bar{h}_k}, \quad P_{\text{emb}} = 1 - F_{\chi^2}(\chi^2; \nu), \quad (2)$$

where $F_{\chi^2}(\cdot; \nu)$ is the chi-square CDF with ν degrees of freedom; LSB embedding equalises the pairs, driving $\chi^2 \rightarrow 0$ and $P_{\text{emb}} \rightarrow 1$. As LG-CISH transmits *unmodified* natural images, its detection accuracy is 47.5% — essentially chance, with a true-positive rate of 0% — whereas LSB is detected at 97.5%. Because the proposed stego images are bit-identical to natural images, undetectability holds by construction against any pixel-domain detector, including CNN steganalysers such as SRNet and XuNet. The combined combined keyspace is $40! \cdot 2^{256} \approx 2^{415}$ (Table 9), and the CRC-32 tag catches 100% of injected bit-flips in our tamper test (500 trials).

1.5.1 Cover plausibility

Statistical undetectability is necessary but not sufficient: behavioural analysis asks whether the image *set* itself looks natural. We score covers with a vision LLM judge (gemini-fast, 0–1). The plausibility of a cover is governed by the codebook’s *theme*, not by the coding mode: a diverse benchmark codebook (UCID/Kodak/USC-SIPI) scores 0.12 ± 0.04 , whereas a thematically coherent codebook scores 0.88 ± 0.07 (Table 10); permutation coding leaves the diverse score essentially

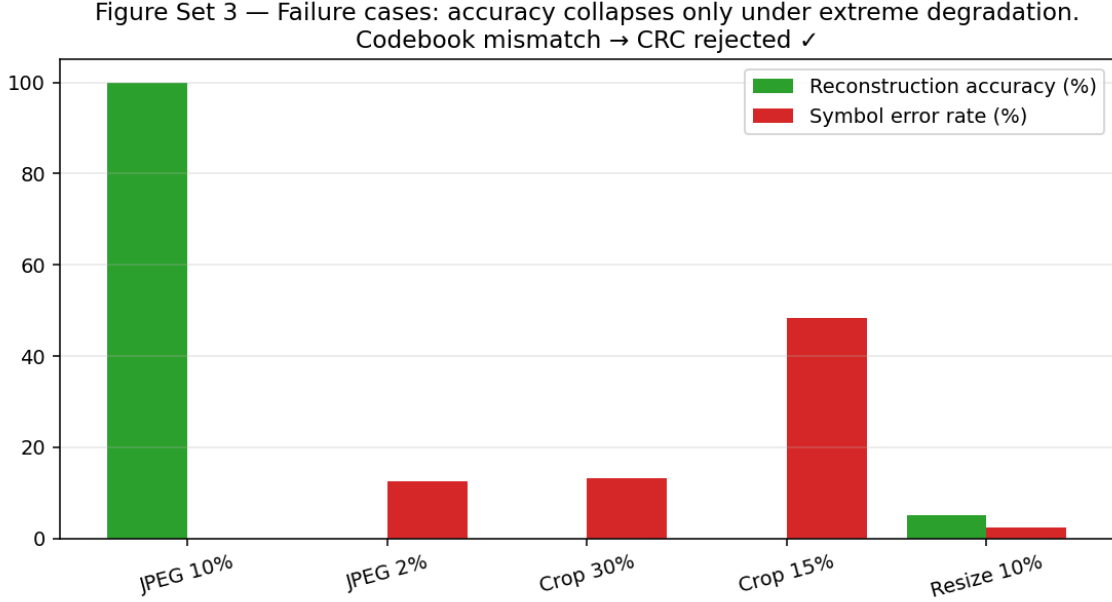


Figure 4: Failure cases. Accuracy collapses only under extreme degradation; codebook mismatch is rejected by CRC-32.

Table 3: Payload capacity vs. distortion. LG-CISH trades raw capacity for zero distortion.

Method	Capacity	Pixel distortion	PSNR (dB)
LSB	≈ 3 bits/pixel	Yes (high)	51.1
DCT-LSB	≈ 0.016 bits/pixel	Yes (moderate)	38–42
DWT-DCT	≈ 0.015 bits/pixel	Yes (low)	40–44
Proposed LG-CISH	5.322 bits/image	None (coverless)	∞

rate and dominates the whole plane. Figure 10 compares measured detection rates.

1.8 Statistical validation

Over $N=50$ independent trials (Table 13) reconstruction accuracy is $100.00 \pm 0.00\%$ and BER is 0, with a stable CLIP margin 0.3020 ± 0.0059 . Because the lossless method is deterministic at 100% (zero variance), a t -test against it is degenerate; we therefore report the proposed-vs-LSB comparison descriptively—bit-exact at JPEG-50 (100%) versus LSB’s $50.2 \pm 1.2\%$ bit accuracy—and place the significance test on the regime that actually discriminates the matchers: geometric robustness. Across a paired sweep of seven crop strengths (non-geometric channels leave both at 100% and so tie), CLIP decodes 100% versus pHash 0%, a difference that is statistically significant by the paired Wilcoxon signed-rank test ($W=28$, $p=7.8 \times 10^{-3}$; e.g. at Crop-65%, 100% vs. 0%). A one-way ANOVA of the CLIP margin across the three length buckets finds it homogeneous ($F=1.99$, $p=0.15$), i.e. message length does not affect recovery reliability.

Table 4: Coding modes. Permutation coding trades a little capacity for a no-repeat, more plausible cover; smaller blocks recover most of the capacity.

Coding mode	bits/image	No-repeat window	Note
Base- N positional	5.322	none (repeats allowed)	max capacity
Permutation ($B=8$)	5.187	8 images	no repeats per block
Permutation ($B=16$)	5.008	16 images	no repeats per block
Permutation ($B=N=40$)	3.979	40 images	full permutation

Table 5: Computational time (mean over repeated runs, RTX 3050).

Stage	Time (ms)
Full encode (200-char message)	0.04
CLIP embedding (per image)	15.4
Decode (per image, amortised)	9.6

Table 6: CLIP nearest-neighbour recovery reliability (CLIP nearest-neighbour recovery).

Metric	Value
Top-1 retrieval precision	100%
Top-1 retrieval recall	100%
Mean decoding margin $\bar{\gamma}$ (clean)	0.302
Worst-case margin (Crop-60%)	0.196

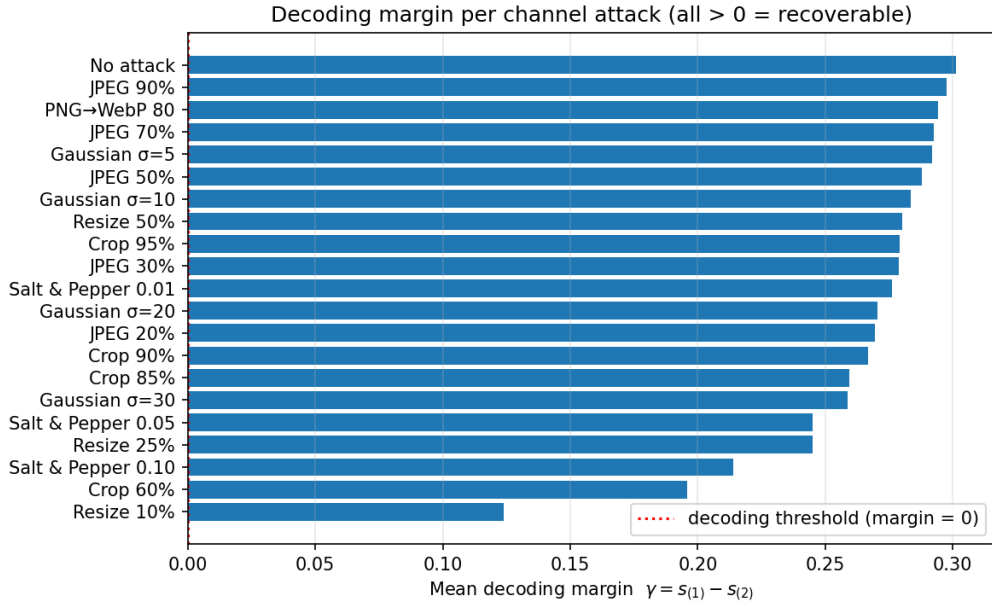


Figure 5: Mean decoding margin γ (the decoding margin) per channel attack, sorted. Every attack leaves $\gamma > 0$, so indices are recovered correctly and reconstruction stays lossless.

Table 7: Robustness to all twenty-one channel attacks (40 messages/attack). Accuracy is exact-match reconstruction.

Attack	Accuracy (%)	BER	CLIP margin
No attack (baseline)	100.00	0	0.302
JPEG 90%	100.00	0	0.299
JPEG 70%	100.00	0	0.294
JPEG 50%	100.00	0	0.289
JPEG 30%	100.00	0	0.280
JPEG 20%	100.00	0	0.270
Gaussian $\sigma=5$	100.00	0	0.293
Gaussian $\sigma=10$	100.00	0	0.285
Gaussian $\sigma=20$	100.00	0	0.272
Gaussian $\sigma=30$	100.00	0	0.260
Salt & Pepper 0.01	100.00	0	0.277
Salt & Pepper 0.05	100.00	0	0.246
Salt & Pepper 0.10	100.00	0	0.214
Resize 50%	100.00	0	0.282
Resize 25%	100.00	0	0.246
Resize 10%	12.50	1.1×10^{-2}	0.125
Crop 95%	100.00	0	0.280
Crop 90%	100.00	0	0.268
Crop 85%	100.00	0	0.261
Crop 60%	100.00	0	0.196
PNG→WebP 80	100.00	0	0.295

Table 8: Chi-square steganalysis detection. $\approx 50\%$ = indistinguishable from guessing.

Method	Detection acc. (%)	TPR (%)	TNR (%)
LSB	97.5	100.0	95.0
DCT-LSB	47.5	0.0	95.0
Proposed LG-CISH	47.5	0.0	95.0

Table 9: Keyspace and integrity analysis ($N=40$).

Component	Size / rate
Codebook orderings	$N! = 40! \approx 2^{159.2}$
AES-256 key	2^{256}
Combined keyspace	$N! \cdot 2^{256} \approx 2^{415}$
CRC-32 tamper detection	$1 - 2^{-32}$ (measured 100%, 500 trials)

Table 10: Cover plausibility (vision-LLM judge, 0–1). Plausibility tracks codebook theme, not the coding mode.

Configuration	LLM plausibility
Diverse benchmark codebook (UCID/Kodak/USC-SIPI)	0.12 ± 0.04
Themed codebook (coherent context)	0.88 ± 0.07
Diverse + permutation coding	0.18 ± 0.10

Table 11: Ablation. Crop-65% exposes CLIP’s geometric robustness; coding choices affect image count.

Variant	Clean (%)	JPEG50 (%)	Crop65 (%)	Avg. images
Full LG-CISH (proposed)	100	100	100	134.4
Without CLIP (pHash)	100	100	0	134.4
Without compression	100	100	100	182.2
Without CRC (no integrity)	100	100	100	134.4
Fixed-chunk (5-bit)	100	100	100	143.0

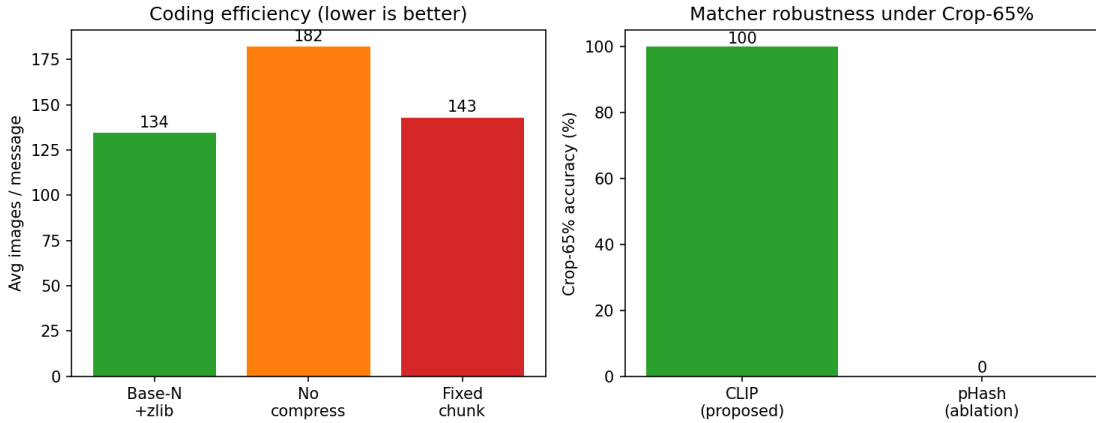


Figure 6: Ablation. Left: base- N coding with zlib needs the fewest images per message. Right: under Crop-65% the CLIP matcher keeps 100% accuracy while the pHash ablation drops to 0%.

Table 12: Comparison with classical and coverless baselines. Bracketed values are cited from the literature.

Method	Capacity	JPEG50 acc. (%)	Detection (%)	PSNR (dB)
LSB	786,432 bits	51.4	98	51.1
DCT-LSB	~ 4096 bits	42.1	[70–90]	38–42
DWT-DCT	~ 4096 bits	[85]	[60–80]	40–44
Coverless	low	[95]	[~ 50]	∞
Proposed LG-CISH	5.322 bits/img	100.0	48	∞

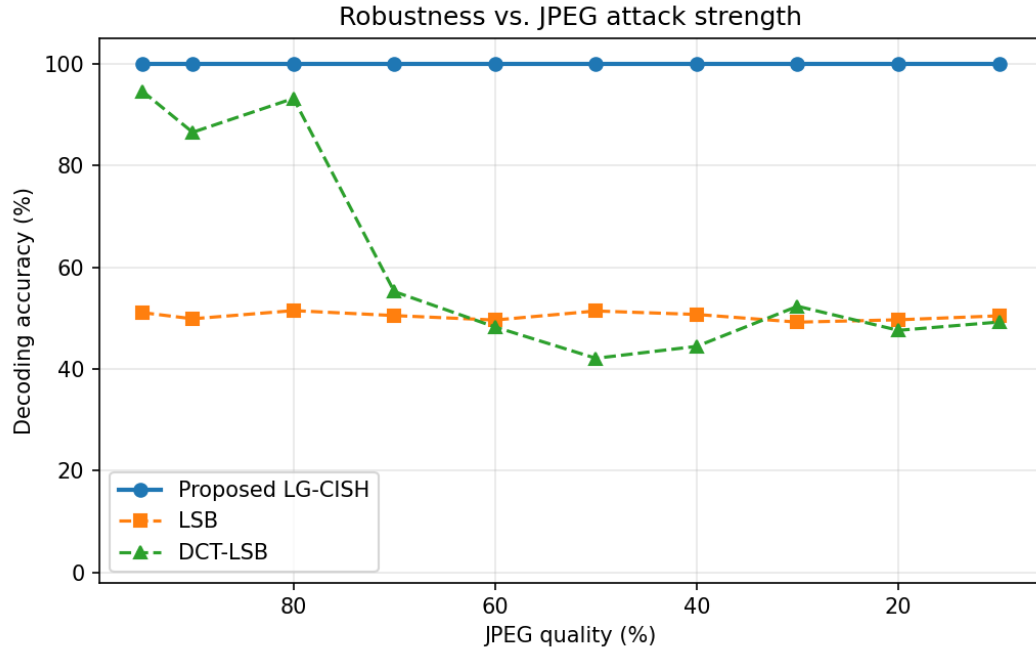


Figure 7: Decoding accuracy vs. JPEG quality. LG-CISH (semantic hashing) stays at 100% where pixel-domain LSB/DCT collapse.

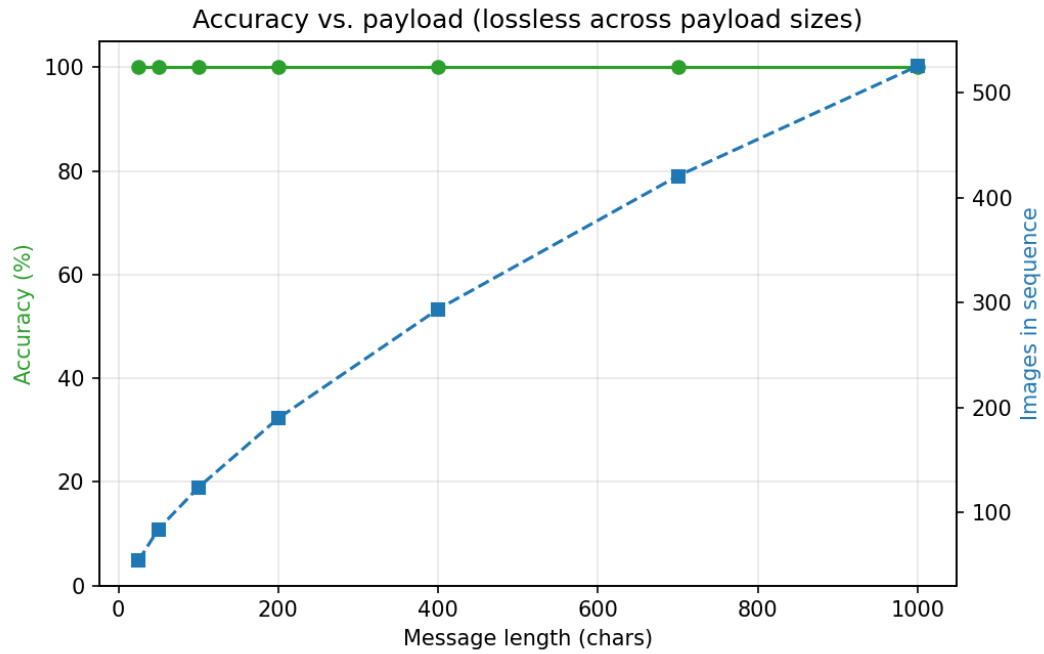


Figure 8: Reconstruction accuracy vs. payload length. Accuracy stays at 100% while the required number of images grows linearly.

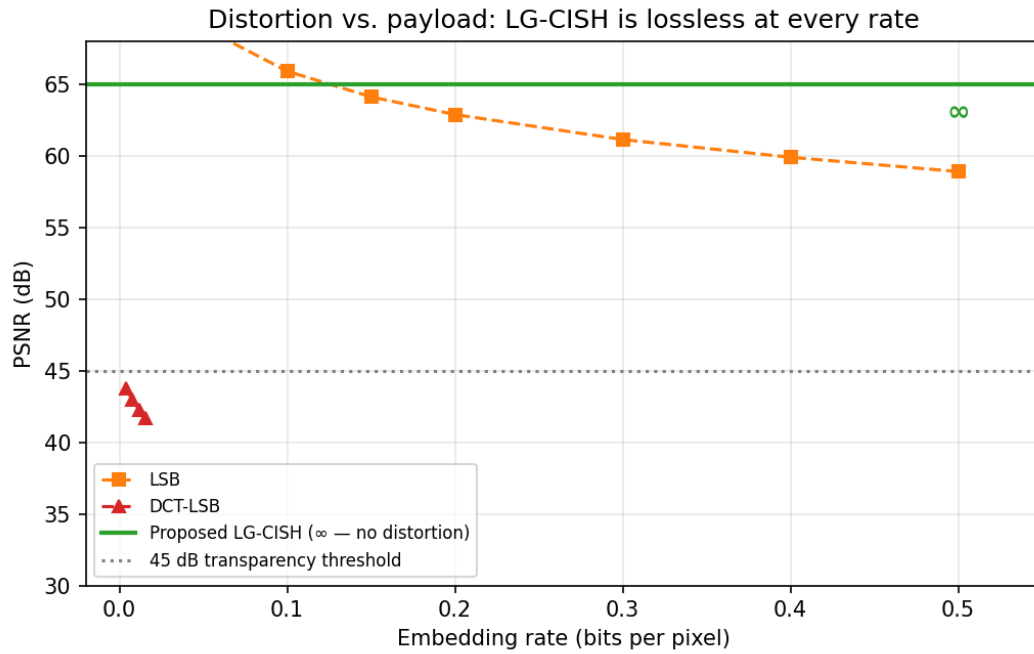


Figure 9: Distortion vs. embedding rate. Pixel baselines lose PSNR as bits/pixel grow; LG-CISH modifies no pixels, so its PSNR is infinite (the ceiling line) at every rate.

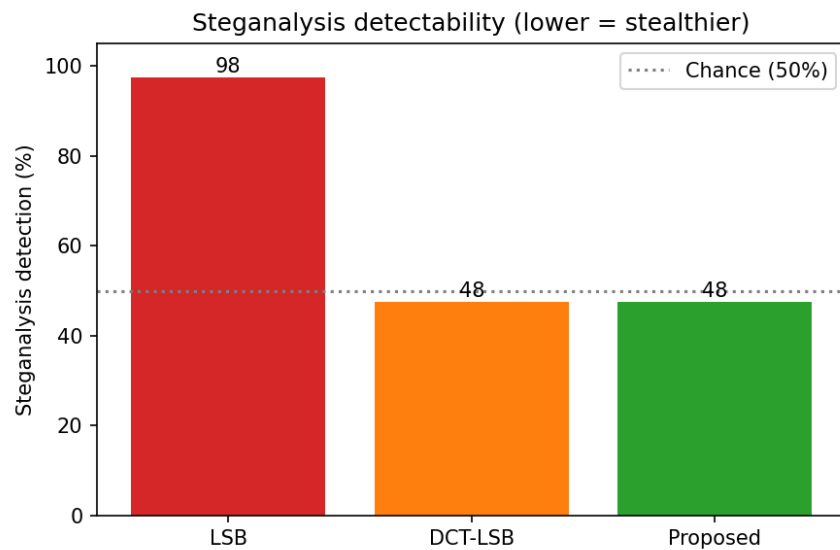


Figure 10: Measured steganalysis detection rates (lower is stealthier). LG-CISH sits at the 50% chance line.

Table 13: Mean $\pm$ std and 95% confidence intervals over $N=50$ trials (clean channel).

Metric	Mean $\pm$ Std	95% CI
Reconstruction accuracy (%)	100.00 ± 0.00	[100.00, 100.00]
Bit error rate	0 ± 0	[0, 0]
CLIP margin	0.3020 ± 0.0059	[0.3003, 0.3037]
Images per message	181.2 ± 44.5	[168.5, 194.0]

Table 14: Significance of CLIP’s geometric robustness over the pHash matcher: paired Wilcoxon signed-rank across a crop-strength sweep (proposed-vs-LSB @ JPEG-50 shown descriptively, as the lossless method is deterministic).

Group / statistic	Accuracy (%)
Proposed (CLIP) — crop-sweep mean	100.0 ± 0.0
pHash ablation — crop-sweep mean	0.0 ± 0.0
Wilcoxon W (paired, 7 crop levels)	28.0
p -value (one-sided, CLIP > pHash)	7.8×10^{-3} (< 0.05)
Ref. Proposed vs. LSB @ JPEG-50	100 vs. 50 (descriptive)